AI Agent Privacy & Safety Checker

Project Proposal Presentation

Abdus Salam

Supervisor: Md. Ziaur Rahman

MBSTU ICT Department

Introduction: The AI Privacy Challenge

Sensitive Data Handling

AI agents process vast amounts of sensitive user data.

New Security Risks

Threats like PII leakage and prompt injection are emerging.

LLM-Specific Issues

Traditional security tools often fail to detect LLM privacy vulnerabilities.

Problem Statement: AI System Vulnerabilities

PII Leakage

Unintended exposure of Personally Identifiable Information.

Prompt Injection

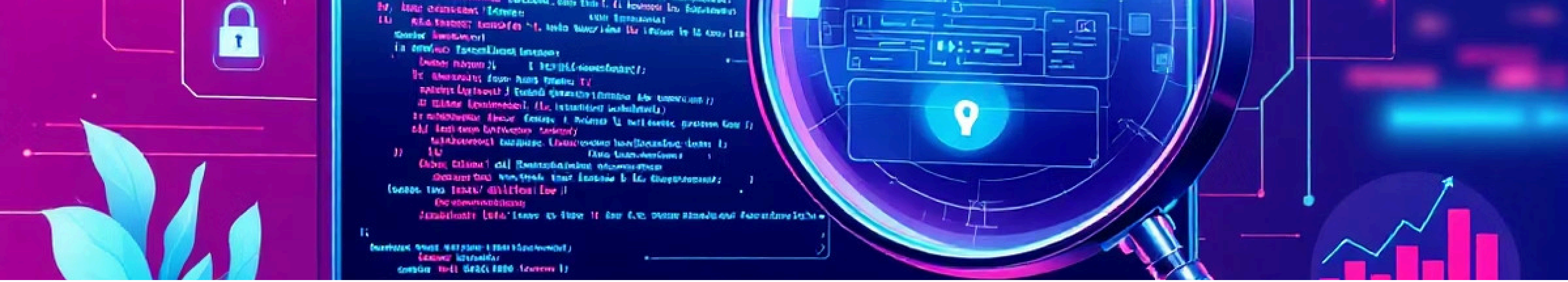
Malicious inputs manipulating AI behavior.

Cross-Session Leaks

Data persistence across user sessions.

Weak Security

Issues with encryption, logging, and third-party telemetry.



Project Goal: Automated AI Privacy & Safety

1

Detect Risks

Identify potential AI privacy risks.

2

Identify Vulnerabilities

Pinpoint specific weaknesses in AI systems.

3

Scan Endpoints

Thoroughly scan LLM and agent endpoints.

4

Generate Reports

Provide actionable insights and compliance checks.

Models Powering Our Checker

spaCy NER

For precise PII detection.

Prompt Injection Test Suite

To detect jailbreak attempts.

Semgrep + Regex

For static privacy code analysis.

MITMProxy Heuristic

To identify network telemetry.

Session-Diff Model

For cross-session leakage detection.

Risk Scoring Model

To determine final severity.

System Architecture Overview

Frontend

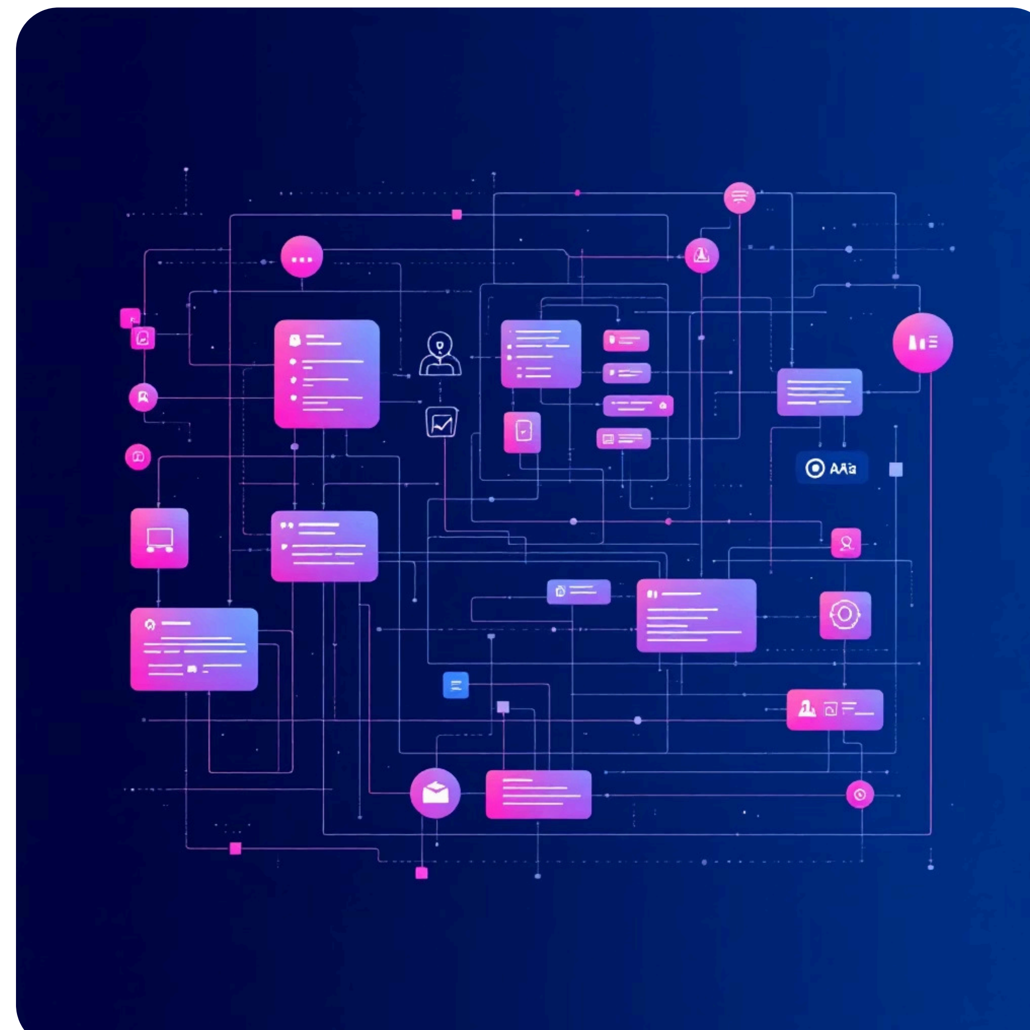
- React
- TypeScript
- Tailwind CSS

Scanner Engine

- Python
- FastAPI
- spaCy NER

Security Modules

- mitmproxy
- Semgrep
- Regex Patterns



Key Features: Comprehensive Protection

PII Leakage Detection

Proactive identification of sensitive data exposure.

Prompt Injection Scanning

Defend against manipulative AI inputs.

Data Retention Checks

Monitor and prevent memory leakage.

Telemetry Detection

Uncover third-party data transmission.

Compliance & Encryption

Ensure logging policies and TLS/Encryption validation.

Diverse Use Cases

1

AI App Developers

Integrate security early in development.

2

Enterprise Data Protection

Safeguard sensitive organizational data.

3

Privacy Auditors

Streamline compliance and risk assessments.

4

Vendor Assessment

Evaluate third-party AI solutions securely.

5

CI/CD Security Testing

Automate security checks in deployment pipelines.

Conclusion: Building Trustworthy AI

Prevent Leaks

Minimizing privacy breaches.

Block Unsafe Deployments

Ensuring secure AI rollouts.

Ensure Compliance

Meeting regulatory standards.

Build Trust

Fostering confidence in AI applications.

Questions & Answers

Thank You